

Proposal: Low-Cost Edge-AI Real-Time Road Infrastructure Auditing

Project Title: RoadWatch

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Hardware Target:

- Edge Inference: **Raspberry Pi 5 (8GB)** with **AI HAT+ (13 TOPS Hailo-8L NPU)**
- Data transmission: LoRa (Long Range on unlicensed ISM bands, Eg: 868 MHz, 915 MHz) modules. For example:
RYLR998 UART Interface 868/915 MHz

1. Executive Summary

This project offers a more efficient, cost effective and autonomous alternative to conventional road damage survey methods, for example: human expeditions costing significant manpower and time, LiDAR vehicles requiring expensive LiDAR sensors and specialized crew.

We propose a distributed edge-AI solution designed for deployment in public transit, municipal vehicles, etc. (vehicles following the AIS 140 mandate) on the widely available SBC (Single Board Computer) **Raspberry pi 5 8gb** variant and an NPU (Neural Processing Unit) **AI HAT+(Hailo-8L NPU)** running a **highly optimized custom trained Yolo v11 CNN(Convolutional Neural Network)** for edge AI inference and data acquisition, **LoRa modules** for v2v (vehicle to vehicle) mesh network for reliable and infrastructure free data delivery. The apparatus once detecting a pothole-saves its coordinates in a local SQLite database, the **LoRa transceiver** looks for peers or the central node to relay the accumulated data via radio waves utilizing a network mesh requiring zero networking infrastructure. This approach provides reliable automated **heatmaps** at a **fraction** of the cost of conventional methods.

2. Problem Statement

A. The Infrastructure Auditing Bottleneck

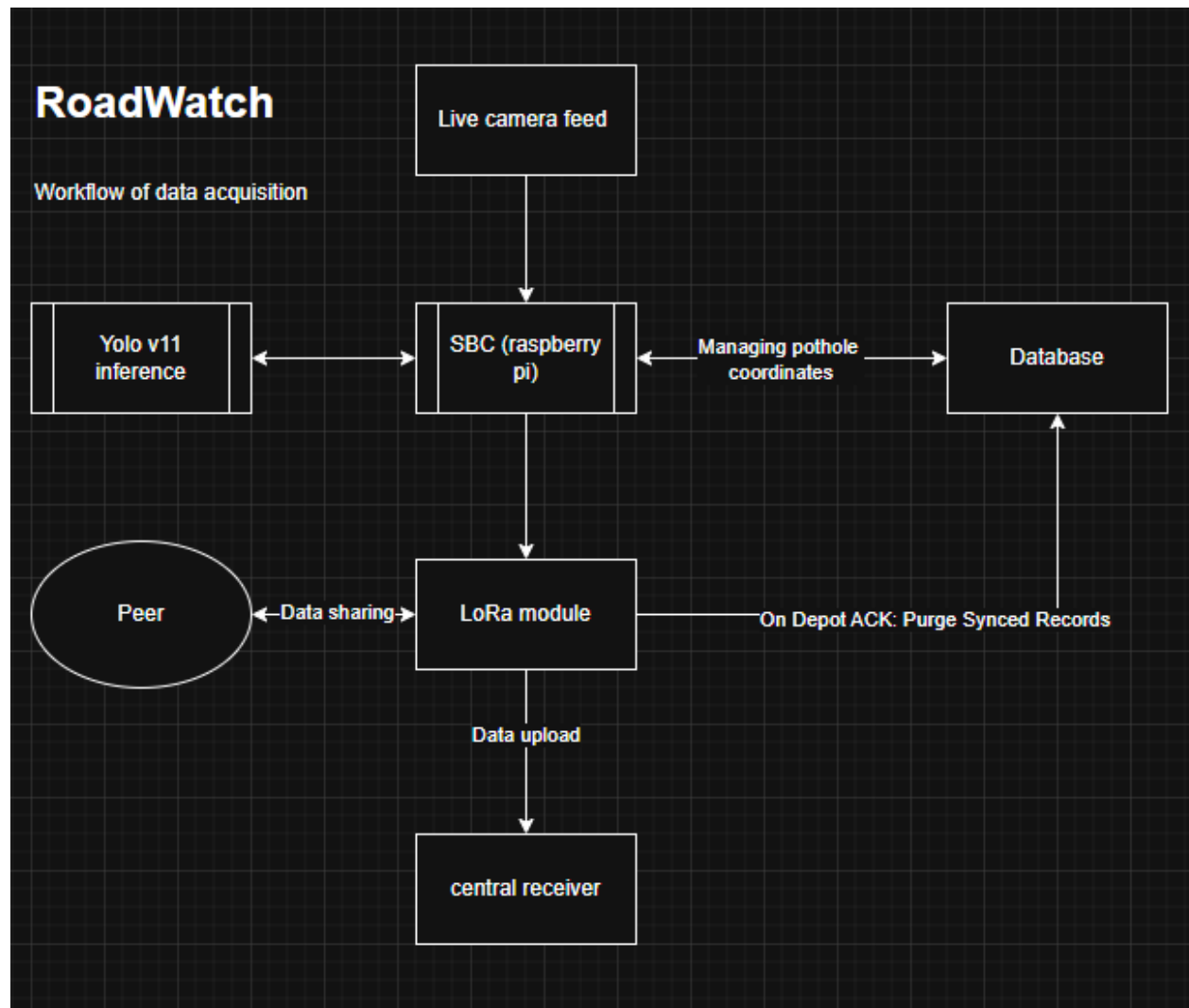
Indian road network spans over 6.3 million kilometers / 3.91 million miles. Current structural monitoring is reactive rather and preventive. Existing automated systems cost (LiDAR) **₹60 Lakhs to ₹95 Lakhs per vehicle** or ongoing commercial survey contracts scaling up to **₹94,000 per Square Kilometer**. Manual visual inspection: engineers or surveyors must physically inspect and log road conditions on standardized charts based on guidelines like **IRC:82** or **ASTM D6433** requiring immense human labor and time and is prone to human errors. And some other non-dominant (in India) methods are utilized mainly on national highways while **manual surveys remain the prominent method for tier-2 and tier-3 urban bodies**.

B. The Bandwidth Bottleneck

Standard cloud-based computer vision pipelines suffer from high latency and prohibitive cellular data costs, making real-time defect logging impossible. Processing must happen **completely on the edge**, outputting only **lightweight metadata payloads** (GPS coordinates, defect class, and confidence score) into a centralized database.

3. Proposed Technology Stack & Methodology

We bypass hardware and bandwidth limits by deploying a highly optimized, single-stage YOLO model running directly on the **local NPU**.

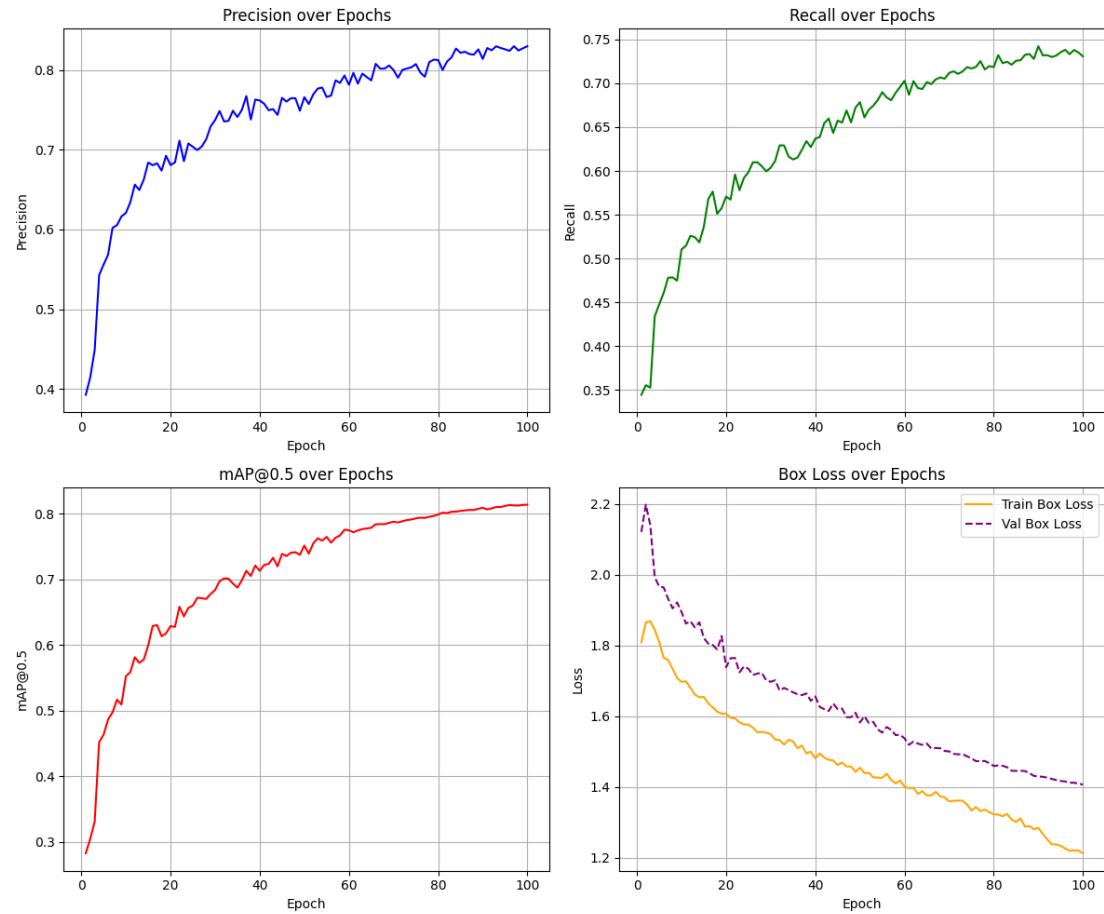


Stage 1: Fast 2D Structural Defect Localizer

- **Baseline Technology:** A specialized, high-precision YOLOv11 model (3.0B parameters) quantized, custom trained and optimized for edge inference.
- **NPU Optimization:** The model is compiled using the Hailo Dataflow Compiler to generate a highly optimized Hailo Executable Format (. hef) file. Quantization from **FP32 to INT8** ensures maximum utilization of the NPU's **13 TOPS** capacity, **yielding low-latency execution at > 30 FPS** with minimal CPU overhead.
- **Software:** the script uses data from the already present **AIS 140 system** to extract real time speed and GPS coordinates. The model uses the live speed to **dynamically** adjust the **inference frequency**, reducing processing load and thermal throttling and redundant data (the model **stops inference completely if the speed is approximately zero**, for instance a traffic jam)
- **Smart data gating:** the script features meteorological gating using an API:

The script utilizes the acquired geological coordinates and plugs them into a weather API to gauge the condition of current weather

- The training metric graph can be seen below:



- **Precision:** 83.0%
- **Recall:** 73.1%
- **mAP@50:** 81.4%
- **mAP@50-95:** 50.9%

Stage 2: Data delivery using mesh topology

- The SOC picks up 6-7 records from the database flagged as “not uploaded” and packs these records into a **compressed comma-separated text block** or **packed binary string** to fit inside the tight hardware limits of the radio link.
- The Raspberry Pi passes the compiled data packet string to the physical LoRa transceiver using an AT command protocol.
- **Half-Duplex RF Broadcast:** The LoRa module shifts from passive listening mode (RX) to transmit mode (TX). It modulates the digital data payload into high-penetration Chirp Spread Spectrum (CSS) radio waves and **broadcasts** it over the legal, **unlicensed 865–867 MHz** Indian band.
- Data parameters held by the packets are **Vehicle_ID, Timestamp, Latitude** and **Longitude**.
- **CSMA (Carrier Sense Multiple Access) protocol (collision detection):**
 - The onboard LoRa module **actively listens for any ongoing transmissions** to not cause interruption and once free the transmission begins. This ensure that in case multiple peers are in **transmission proximity** every node does not start transmitting as to not “**clog**” the frequency which could **corrupt the data**.
- **Checksum Verification:** The **receiving node** (either a passing peer bus or the fixed depot gateway antenna) captures the incoming radio waves. Its internal hardware validates the packet's **CRC (Cyclic Redundancy Check)** to ensure zero bits were dropped or corrupted during transit. Malformed packets are silently discarded at the hardware level.
- If the **depo central receiver** gets the packet it flags it as synched and the edge device clears the database.
- **Finally:** the server compiles all the data on a **per-day basis** and discards redundant entries using the vehicle id, timestamps and coordinates (in the case a peer shared the same data to multiple other peers) and maps the coordinates on a **Django based website** for users.

4. Current Development Status & Technical Feasibility

Our pipeline is past the conceptual stage and is highly functional:

1. **Model Baseline:** Our high-performance model demonstrates robust classification performance under varying illumination and pavement conditions.
2. **Inference Scripting:** Completed a high-efficiency runtime environment (`runtimeTest.py`) supporting real-time camera feeds and desktop screen-capture loops using accelerated inference libraries (test footage is linked in sources).
3. **Data Preprocessing Pipelines:** Created automated toolsets (`convert_dataset.py`) to clean, filter, and normalize raw road defect data, dropping "noise" parameters like paint-wear blur to prevent false positives in deployment.

5. Project Milestones & Timeline

a. Milestone 1: Celery Integration for Scheduled Spatial Compilations

- Deploy Celery backed by a Redis broker, to handle all long-running, asynchronous database chores off the main web thread.
- Configure a Celery Beat scheduler to trigger the spatial-temporal deduplication algorithm periodically (removing the redundant entries). e.g., every 120 minutes.
- Task Logic: Automatically fetch fresh raw ingest arrays from the depot gateway database, run the 5-meter spatial radius cluster evaluation or heat-based road highlights, map them into clean GeoJSON layers, and clear out the raw temporary tables.

b. Milestone 2: Server Code Optimization & UI Polish

- Optimize SQL query bottlenecks by indexing location coordinates (Latitude, Longitude) and Unix timestamps.
- Polish the frontend map interface (Mapbox/OpenStreetMap) to cleanly render real-time repair tickets and cluster markers without lagging during massive point loads.

c. Milestone 3: CSMA Firmware & Handshake Logic Optimization

- Write the Non-Persistent CSMA protocol using live RSSI sniffing routines via the RYLR998 AT commands.
- Optimize the randomized exponential backoff timing windows to ensure high packet-delivery rates when multiple nodes compete for the 866 MHz channel.
- Verify structural SQLite transactions (DELETE on matching depot ACK) to ensure memory consumption on the micro-SD card remains completely constant.

And more....

6. Budget & Resource Allocation (Phased Funding Model)

- Edge hardware Bill Of Materials (per vehicle)

Component	Technical Specifications	Cost (INR)
SBC Host	Raspberry Pi 5 (8GB RAM, Quad-Core @ 2.4GHz)	₹7,500
Edge Coprocessor	Raspberry Pi AI HAT+ (13 TOPS Hailo-8L NPU via PCIe Gen 2.0)	₹3,500
Transceiver	REYAX RYLR998 LoRa Module (865-867MHz UART Serial Interface)	₹900
Optical Sensor	1080p/60fps Wide-Angle USB Dashboard Camera	₹1,500
GPS / Telemetry Module	AIS 140 Compliant USB GPS/IMU Receiver	₹1,200
Enclosure & Power	Custom Rugged Enclosure + 5V/5A Active Cooler PD Unit	₹1,000
Total Hardware Cost	Per-Vehicle Edge-Node Asset Footprint	₹15,600

- Operational & Server Infrastructure (12-Month Pilot Phase)

Operational Unit	Description / Infrastructure Stack	Cost (INR)
Depot Gateway Node	Dedicated Raspberry Pi 5 + High-Gain Outdoor Omni Antenna for LoRa receiving	₹8,500
Web Host & API Compute	Virtual Private Server (VPS) running Django/FastAPI & Gunicorn	₹24,000
Asynchronous Task Queue	Managed Redis Broker + Celery Worker Node (for spatial deduplication engine)	₹18,000
Telemetry Database	Managed PostgreSQL Instance with PostGIS spatial indexing extension	₹12,000
GIS Mapping API Allocation	Mapbox/OpenStreetMap tile API routing and vector tile allocation	₹10,000
Domain & Security Suite	SSL Certificates, Domain Registration, and Cloudflare DDoS protection	₹7,500
Total Server Infrastructure	Annual Cloud, Web, & Gateway Operating Cost	₹80,000

- **Consolidated Phase 1 Grant Budget (TIDE 2.0 Limit: ₹5,00,000)**

Category	Description / Deliverable Breakdown	Cost (INR)
Edge Hardware (4 Nodes)	4x Raspberry Pi 5, Hailo-8L NPUs, LoRa modules, and accessories	₹63,000
Field Equipment & Mounts	Ruggedized vehicle mounting rigs, wiring, and custom enclosures for 4 nodes	₹20,000
Server & Cloud Infrastructure	12-Month database hosting, Celery/Redis queue compute, and GIS API licenses	₹80,000
Travel, Fuel & Calibration	Vehicle fuel, testing toll allowances, and on-road sensor-fusion debugging runs	₹50,000
Technical Manpower	1x Systems Integration Engineer (Contract for R&D, sensor calibration, and integration)	₹2,87,000
TOTAL REQUESTED BUDGET		₹5,00,000

7. Expected Impact and Commercial Viability

- **Safety:** Early detection of potholes and alligator cracking on urban expressways prevents severe high-speed accidents.
- **Resource Optimization:** Delhi municipal bodies (MCD, NDMC) can transition from reactive patch-working to predictive resurfacing, lowering maintenance costs by up to 40%.
- **Accessibility & Legal Handoff:** The technology is built on a scalable hardware layout (under ₹15,000 per vehicle). Incubating at FITT IIT Delhi ensures clean transition from an academic prototype to an incorporated commercial startup.

8. Sources

- **DATASET:** <https://universe.roboflow.com/smartathon/new-pothole-detection>
- **Inference demo**(running on Ryzen 5500H) on a video: <https://youtu.be/TaBGau3z3X0>
- **Apparatus demo** (running on raspberry pi without the NPU and using laptop to host a location server as a placeholder for AIS 140): <https://youtu.be/6Rtu6VWgYoY>
- **Modal Framework:** <https://docs.ultralytics.com/>

